

ADVERTISEMENT

NumPy ufuncs

[< Previous](#)[Next >](#)

What are ufuncs?

ufuncs stands for "Universal Functions" and they are NumPy functions that operate on the `ndarray` object.

Why use ufuncs?

ufuncs are used to implement *vectorization* in NumPy which is way faster than iterating over elements.

They also provide broadcasting and additional methods like `reduce`, `accumulate` etc. that are very helpful for computation.

ufuncs also take additional arguments, like:

`where` boolean array or condition defining where the operations should take place.

[Tutorials](#) ▼[References](#) ▼[Exercises](#) ▼[Certificates](#) ▼[Sign In](#)[HTML](#) [CSS](#) [JAVASCRIPT](#) [SQL](#) [PYTHON](#) [JAVA](#) [PHP](#) [HOW TO](#) [W3.CSS](#) [C](#) [C++](#) [C#](#) [BOOT!](#) >[Data Distribution](#)[Random Permutation](#)[Seaborn Module](#)[Normal Distribution](#)[Binomial Distribution](#)[Poisson Distribution](#)[Uniform Distribution](#)[Logistic Distribution](#)[Multinomial Distribution](#)[Exponential Distribution](#)[Chi Square Distribution](#)[Rayleigh Distribution](#)

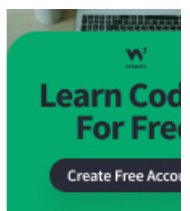
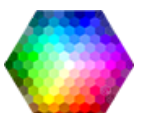
What is Vectorization?

Converting iterative statements into a vector based operation is called vectorization.

It is faster as modern CPUs are optimized for such operations.

Add the Elements of Two Lists

list 1: [1, 2, 3, 4]

**COLOR
PICKER**

[Pareto Distribution](#)[Zipf Distribution](#)

NumPy ufunc

[ufunc Intro](#)[ufunc Create Function](#)[ufunc Simple Arithmetic](#)[ufunc Rounding Decimals](#)[ufunc Logs](#)[ufunc Summations](#)[ufunc Products](#)[ufunc Differences](#)[ufunc Finding LCM](#)[ufunc Finding GCD](#)[ufunc Trigonometric](#)[ufunc Hyperbolic](#)[ufunc Set Operations](#)

Quiz/Exercises

[NumPy Editor](#)[NumPy Quiz](#)

list 2: [4, 5, 6, 7]

One way of doing it is to iterate over both of the lists and then sum each elements.

Example

[Get your own Python Server](#)

Without ufunc, we can use Python's built-in `zip()` method:

```
x = [1, 2, 3, 4]
y = [4, 5, 6, 7]
z = []

for i, j in zip(x, y):
    z.append(i + j)
print(z)
```

[Try it Yourself »](#)

NumPy has a ufunc for this, called `add(x, y)` that will produce the same result.

Example

With ufunc, we can use the `add()` function:

```
import numpy as np

x = [1, 2, 3, 4]
y = [4, 5, 6, 7]
z = np.add(x, y)

print(z)
```

[Try it Yourself »](#)

Exercise ?

What does 'ufuncs' stand for?

- ☐ United Functions
- ☐ Universal Functions



AD

R